Air Quality and Income Inequality: Canadian Metropolitan Areas

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Your abstract here.

As climate change reshapes environmental regimes worldwide, academic and political interest in the interactions between climatic and social systems has grown. Environmental exposures, including air pollution, are understood to impact human outcomes (Burke et al., 2022). Studies which evaluate the varied effects of environmental exposures across social and economic subgroups are especially salient (Landrigan et al., 2018). However, the impact of environmental exposures on urban economic conditions is underexplored.

Air pollution is a particularly important source of environmental exposure: globally, "approximately 86% (2.5 billion inhabitants) of urban inhabitants lived in urban areas that exceeded WHO's 2005 guideline annual average $PM_{2.5}$" (Southerland et al., 2022). Previous research has determined that stringent air quality regulations, such as those instated by the Clean Air Act, can exacerbate income inequality (Jha et al., 2019). This conclusion may cause regulatory agencies to avoid implementing environmental policy in order to avoid negative economic outcomes. Under the Clean Air Act, a county that is out of attainment with the American National Ambient Air Quality Standards is subject to "stricter environmental regulations than attainment counties" (Jha et al., 2019).

This detail leads to a potential misattribution of causality in the relationship between strict air quality standards and income inequality. If urban areas with poor air quality are less resilient to increases in income inequality, it could be poor air quality—not air quality regulation—which is associated with higher relative income inequality. In principle, this destabilizing effect is possible. Poor air quality has a disproportionately negative impact on low-income households (Deryugina et al., 2021). Additionally, poor air quality is associated with worsened labor productivity due to adverse health impacts (Office of Air and Radiation, 2011). Together, these impacts on the workforce might worsen or prolong the short- and long-term impacts of an economic shock on income inequality.

A first step to determining whether air quality has a destabilizing effect on income equality is to determine whether a relationship between air quality and income inequality exists. **This paper asks if air quality has a statistically significant effect on income inequality within Canadian Metropolitan Areas (CMAs)**. If the social consequences of poor air quality worsen an economy's resilience to income inequality, we would find higher income inequality in CMAs with generally higher air pollution. As such, **we expect to find a positive relationship between air pollution and income inequality**. By illustrating this relationship, we hope to contribute to a robust understanding of the connection between environmental exposures and human outcomes. Assessing the complex interactions between climatic and social systems allows for the implementation of more accurate, equitable, and effective environmental policy. !ADD OVERVIEW OF KEY FINDINGS

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I. Data Description

All of the data used in this research was collected from one of three sources: Environment and Climate Change Canada, Statistics Canada's 2021 Census, and Environics Analytics' 2024 Demographic Estimates & Projections.

Air Quality Health Index (AQHI) is a metric of air pollution created by Environment and Climate Change Canada. We have chosen CMA average AQHI as our independent variable of interest. Using a risk-weighted scale of ground-level ozone (O_3), particulate matter ($PM_{2.5}$ & PM_{10}) and nitrogen dioxide (NO_2), the index provides a value typically ranging between 1 and 10 (Government of Canada, 2021). **Higher AQHI values indicate higher air pollution.** AQHI values between 1 and 3 are associated with low health risk; values between 4 and 6 are associated with a moderate health risk; values between 7 and 10 are associated with a high health risk; and values above 10 indicate a very high health risk (Government of Canada, 2015). The AQHI index is typically reported to the public as integer values—but decimal hourly observation data from December 2023 until April 2025 is currently available. Hourly data is organized into CSV files by month of observation and geographic region. Due to provincial policy, no AQHI data from Quebec is publicly available (Environment and Climate Change Canada, 2025). We collected all AQHI observations from 2024 and combined each measurement station's hourly data into a daily average AQHI. We also collected a GeoJSON file which contains spatial data about every AQHI measurement station (Environment and Climate Change Canada, 2025). This GeoJSON file was converted to the SHP file format.

To associate this AQHI data with corresponding CMAs, we collected provincial and CMA spatial data files from Statistics Canada's 2021 Census data (Statistics Canada, 2021a; Statistics Canada, 2021b). This data is a set of boundary measurement SHP files which were last updated in 2021. This is acceptable: CMA boundaries have not changed since the 2021 census. In addition to providing geographic data, these boundary sets include a land area measurement (in km^2) and Census Population Designation for each CMA. All CMAs studied in this paper have one of three Census Population Designations: "B," a CMA with a population >100,000 with >50,000 living in a designated CMA core ($n = 34$); "D," an untracted Census Agglomeration with a designated core population >10,000 ($n = 23$); and "K," a tracted Census Agglomeration with a designated core population >50,000 ($n = 6$) (Statistics Canada, 2018). Once we adjusted the Coordinate Reference Systems of all relevant geographic data, we were able to group CMAs with their AQHI measurement stations. Not all AQHI measurement stations are located within a CMA, and not all CMAs contain an AQHI measurement station—these incomplete measurements are excluded from our data. Additionally, when a CMA contained multiple AQHI measurement stations, their daily averages were combined. Having organized our daily average AQHI data by CMA, we then calculated each CMA's average AQHI in 2024. Additionally, we calculated each CMA's percentage of daily average AQHI values greater than 3. Days without any AQHI reading were omitted from the calculation of these averages. One CMA, Brandon, was ommitted because it had less than 100 days of collected AQHI data: this could introduce a potential bias in AQHI measurement, as air pollution varies across seasons (Zareba and Weglinska, 2024).

Demographic data about every CMA was collected from Environics Analytics' 2024 Demographic Estimates & Projections (Environics Analytics, 2024). Official census data for 2024 does not exist, but this dataset—which was created "in conjunction with experts at The Centre for Spatial Economics"—is one of the best available projections of Canadian demographic data (Environics Analytics, 2024). It was derived from Statistics Canada's 2021 Census data and made available through the SimplyAnalytics platform. We chose to collect the following 2024 data about every CMA:

1) Mean Income ($)

2) Median Household Income ($)
3) Percent of Households with a Household Income $100,000 or Over (%)
4) Percent of Households with a Household Income of $100,000 or Over with a Household Income of $200,000 or Over (%)
5) Percent of Households with a Household Income of $200,000 or Over with a Household Income of $300,000 or Over (%)
6) Percent Male Population (%)
7) Percent of Population with Visible Minority Status (%)
8) Percent of Population >15 Years Old Married or Living With A Common-Law Partner (%)
9) Percent of Population without Knowledge of English or French (%)
10) Percent Immigrant Population (%)
11) Percent of Population >15 Years Old in the Labour Force (%)
12) Employment Rate (%)

The CMA demographic data does not include a measurement of income inequality. The Gini Coefficient is a measure of income inequality $Gini \in [0, 1]$, with $Gini = 0$ representing perfect income equality and $Gini = 1$ representing maximum income inequality. We have chosen the Gini Coefficient as our dependent variable. We estimated the Gini Coefficient for each CMA population using the following vector $\vec{I_i}$ of 2024 demographic data:

$$\vec{I_i} = \begin{bmatrix} I_0\text{: Median Household Income} \\ I_1\text{: Percent of Households with a Household Income \$100,000 or Over} \\ I_2\text{: Percent of Households with a Household Income \$100,000 or Over with a Household Income of \$200,000 or Over} \\ I_3\text{: Percent of Households with a Household Income \$200,000 or Over with a Household Income of \$300,000 or Over} \end{bmatrix}$$

From $\vec{I_i}$ we can approximate the Lorenz Curve and corresponding Gini Coefficient of every CMA. We used a monotonically increasing cubic spline interpolation to generate a polynomial representing the Lorenz Curve of every CMA. The Gini Coefficient was then approximated through numerical integration. The complete approximation process is detailed in Appendix A.

Two CMAs were excluded from the dataset because of their estimated Gini Coefficients: Wood Buffalo and Yellowknife. Both CMAs have very high mean incomes, which means that our upper bound of household income ($300,000) is not sufficient to accurately estimate the concavity of the Lorenz Curve.

No noticable outliers are present in the remainder of our CMA demographic data. All other listed variables acted as control variables, and their purpose will be discussed further in our Model Specification section.

II. Summary Statistics

A map of all relevant AQHI measurement stations and CMAs is available in Appendix B.

Table 1 provides descriptive statistics for the primary variables employed in our empirical analysis. The table summarizes data from CMAs across Canada, including measures of income inequality, air quality, demographic characteristics, and CMA land area. The mean Gini Coefficient is 0.383 with a standard deviation of 0.100, reflecting variation in income inequality across CMAs. The

mean average AQHI is 2.054 ($s = 0.287$), and the mean percentage of high AQHI days is 19.44% (s = 21.932).

Table 1—: Census Metropolitan Areas: Key Variables ($n = 63$)

Variable	**Mean**	**S.D.**	**Min.**	**Max.**
Gini Coefficient	0.383	0.100	0.142	0.586
Average AQHI	2.054	0.287	1.455	2.689
High AQHI Days (%)	19.444	21.932	0.000	85.000
Average Income ($)	56,564.91	7,484.71	43,216.52	77,301.47
Male Population (%)	49.713	0.854	48.156	52.114
Visible Minority (%)	15.394	12.093	2.208	58.503
Married or Common-law (%)	56.331	2.444	48.760	61.884
No English or French (%)	0.902	1.091	0.074	5.639
Immigrant (%)	15.888	9.435	3.017	49.748
Labour Force (%)	51.357	4.267	42.844	62.069
Employment Rate (%)	91.359	2.212	82.300	96.600
Land Area (km^2)	3491.412	4223.093	54.843	21708.620

Note: S.D. represents standard deviation. Min. represents the minimum value of a variable within our CMA data. Max. represents the maximum value of a variable within our CMA data.
Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

It is helpful to understand the simple correlation between each of these variables: high correlation values could be an indicator of multicollinearity. Table 2 displays the correlation between all considered variables.

We began by estimating a simple linear regression of income inequality (Gini Coefficient) on average AQHI, as displayed in Figure 1.

III. Model Specification

A. Main Model Specification

The core objective of this model is to estimate the effect of air quality (as captured by AQHI) on income inequality (as measured by the Gini Coefficient) within CMAs. To investigate the relationship between air quality and income inequality, we employ an ordinary least squares (OLS) regression model:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i \times IncomeCat_i + \sum_{k=1}^{6} (\beta_{k+2} W_{k,i}) + \epsilon_i \tag{1}$$

The dependent variable in this model is $Gini_i$, our estimation of the Gini Coefficient. The independent variable of interest is $AQHI_i$, which is the average AQHI in a given CMA. This captures typical air quality conditions, accounting for various types of pollutants (Government of Canada, 2021). This model will directly test whether *ceteris paribus* differences in air quality are associated with changes in the Gini Coefficient. For example, if the estimated coefficient β_1 on $AQHI_i$ is significant and positive, that would indicate that worse air quality is correlated with higher income inequality within a CMA. We also have included an interaction variable constructed from

Table 2—: Census Metropolitan Areas: Variable Correlation ($n = 63$)

Variable	G.C.	A.A.	H.A.D.	A.I.	M.P.	V.M.	M.C.	N.E.F.	I.	L.F.	E.R.	L.A.
Gini Coefficient	—	-0.224	-0.267	-0.879	-0.551	-0.475	-0.277	-0.302	-0.527	-0.726	-0.397	-0.018
Average AQHI		—	0.907	0.377	0.237	0.359	0.019	0.212	0.361	0.230	-0.023	0.004
High AQHI Days (%)			—	0.407	0.288	0.393	0.085	0.233	0.431	0.244	-0.009	0.137
Average Income ($)				—	0.471	0.360	0.329	0.114	0.405	0.650	0.363	0.098
Male Population (%)					—	0.192	0.240	0.178	0.185	0.461	0.184	0.189
Visible Minority (%)						—	0.042	0.880	0.971	0.550	0.135	-0.027
Married or Common-law (%)							—	0.006	0.083	0.362	0.324	0.034
No English or French (%)								—	0.835	0.354	0.089	-0.051
Immigrant (%)									—	0.569	0.232	-0.047
Labour Force (%)										—	0.420	-0.032
Employment Rate (%)											—	-0.153
Land Area (km^2)												—

Note: All values listed are the simple correlation $\rho_{X,Y}$ of the associated row and column variables. The correlation matrix is symmetric. G.C. represents Gini Coefficient. A.A. represents Average AQHI. H.A.D. represents High AQHI Days (%). A.I. represents Average Income ($). M.P. represents Male Population (%). V.M. represents Visible Minority (%). M.C. represents Married or Common-law (%). N.E.F represents No English or French (%). I. represents Immigrant (%). L.F. represents Labour Force (%). E.R. represents Employment Rate (%). L.A. represents Land Area (km^2).
Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

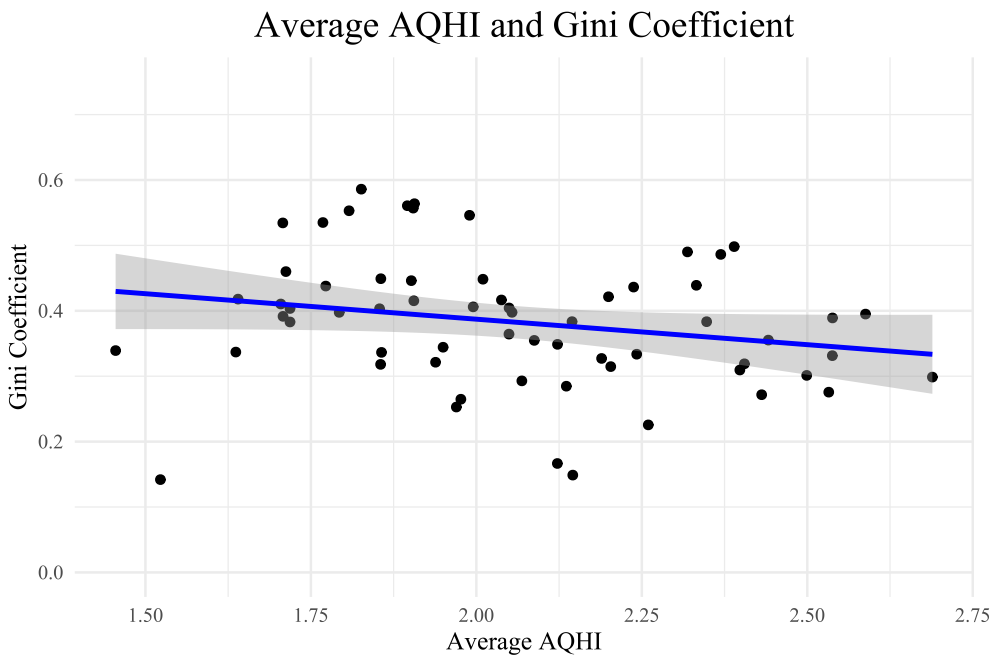


Figure 1. : Scatterplot and Bivariate Regression of CMA Average AQHI and Gini Coefficient

Note: Scatterplot represents CMA data values for Average AQHI and Gini Coefficient. The blue line represents the simple bivariate regression of the form $\hat{Y} = \beta_0 + \beta_1 X_i$ where $\hat{Y}$ represents predicted Gini Coefficient, β_0 represents the predicted intercept, β_1 represents the coefficient on Average AQHI, and X_i represents a compact space of Average AQHI values such that $X_i \in [1.455, 2.689]$: the range of our collected values of Average AQHI. β_0 had an estimated value of 0.543 with a standard error of 0.090. β_1 had an estimated value of -0.078 with a standard error of 0.043.
Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

$AQHI_i \times IncomeCat_i$, which is a categorical variable representing the mean income for a given CMA. We include this variable because we expect air quality to effect income inequality differently in places with higher or lower average incomes due to the heterogenous health impacts of air pollution (Deryugina et. al, 2021). Adding in this interaction variable will not introduce endogeneity: mean income was not used in our calculation of the Gini Coefficient, and the interaction variable only accounts for changes in the effect of $AQHI_i$ on $Gini$ across different levels of average CMA income. Since we are not including average income directly, we have centered and categorized CMA mean incomes to make our interpretation of the model more clear and avoid any problems caused by potential non-linearity in the interaction variable.

We have chosen the following control variables:

1) Percent Male Population (%)
2) Percent of Population >15 Years Old Married or Living With A Common-Law Partner (%)
3) Percent Immigrant Population (%)
4) Percent of Population >15 Years Old in the Labour Force (%)
5) Employment Rate (%)
6) CMA Land Area (km^2)

Percent male population is included as a control variable because there is higher male representation in industries like manufacturing, which are correlated with both air pollution and income inequality (Varechova et al., 2016).

Marital status is included as a control variable: in lower-income regions, dual-income households often exhibit higher financial stability than single-income households (Pais et al., 2014).

Percent immigrant population is included as a control variable because recent immigrants tend to concentrate in low-income, high-pollution areas where income inequality is often the greatest (American Lung Association, 2023).

We include labor force participation rate and employment rate to adjust for income polarization and CMA variations in economic prosperity during 2024.

CMA land area is included as a control variable to address potential heterogeneity between CMAs caused by geographic fixed effects. Geographically larger CMAs might vary in both pollution levels and income inequality; the inclusion of CMA land area as a control variable adjusts for this fixed effect.

We ultimately decided to omit two control variables to account for high multicollinearity. The percentage of immigrants in total population has an extremely high correlation with both the percent of population with visible minority status ($\rho = 0.971$) and the percent of population without knowledge of English or French ($\rho = 0.835$). As such, both of these variables were not included in our regression.

The error term represents unobserved factors influencing local income inequality that are not explicitly included in the model.

B. Model Variations

To ensure the validity of our results, we propose four alternative models.

We first add $AQHI_i^2$ to our regression to account for a possible nonlinear effect of air quality on income inequality. Different levels of environmental exposure to poor air quality might be associated with varying health and economic consequences: by including both a linear and a quadratic term, we can test whether the effects of higher pollutant concentrations follow a simple linear pattern or exhibit diminishing/increasing marginal effects (Banzhaf & Walsh, 2008; Landrigan et. al, 2018).

This gives us the following alternative functional form:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i^2 + \beta_3 AQHI_i \times IncomeCat_i + \sum_{k=1}^{6} (\beta_{k+3} W_{k,i}) + \epsilon_i \quad (2)$$

Next, we replace the interaction variable $AQHI_i \times IncomeCat_i$ with the quadratic term $AQHI_i^2$. This change tests whether AQHI's effect on income inequality actually varies in CMAs with higher or lower average incomes: it's possible that a direct nonlinear effect of air quality on inequality is obscured by multicollinearity with the interaction variable. This gives us an alternative functional form:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i^2 + \sum_{k=1}^{6} (\beta_{k+2} W_{k,i}) + \epsilon_i \quad (3)$$

Next, we remove the control variable for CMA land area to verify that our results are not unduly influenced by the inclusion of geographic fixed effects and that we are not overfitting our regression. This gives us the following alternative functional form:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i \times IncomeCat_i + \sum_{k=1}^{5} (\beta_{k+2} W_{k,i}) + \epsilon_i \quad (4)$$

Finally, we group our data by CMA Census Population Designation to capture potential heterogeneity caused by population size: more populous CMAs may vary in both pollution levels and income inequality. This gives us a regression model of the functional form:

$$Gini_i = \beta_0 + \beta_1 AQHI_i + \beta_2 AQHI_i \times IncomeCat_i + \sum_{k=1}^{6} (\beta_{k+2} W_{k,i}) + \epsilon_i \quad (5)$$

IV. Table of Results

Table 3 displays the results of our main regression and five alternative model specifications. Significance values of each coefficient estimate are noted.

V. Discussion

Our findings suggest that air quality is a significant determinant of income inequality, but its impact is highly dependent on regional income levels. In lower-income regions, poorer air quality is associated with higher income inequality. This aligns with environmental justice literature, which often finds that pollution burdens fall disproportionately on marginalized and economically vulnerable populations.

Interestingly, in higher-income regions, increased AQHI is associated with lower inequality. One possible explanation is that wealthier areas have more robust public health infrastructures or adaptive capacities that mitigate the harmful distributional effects of air pollution.

Additionally, demographic variables like the male population share and visible minority status played notable roles. A higher percentage of male population was associated with lower inequality, while a greater visible minority population slightly increased it, though with weaker significance.

These insights could help inform policy targeting environmental and social inequality. In particular, improving air quality in lower-income CMAs could have a meaningful impact on reducing

Table 3—: Main vs. Robustness Checks on Determinants of Income Inequality Across Canadian CMAs ($n = 63$)

	Dependent Variable: Gini Coefficient				
	Main	Alt. 1	Alt. 2	Alt. 3	Alt. 4
Equation Number	(1)	(2)	(3)	(4)	(5)
Intercept	2.807***	3.424***	3.591***	2.766***	2.791***
	(0.319)	(0.459)	(0.701)	(0.343)	(0.325)
Average AQHI	0.089**	−0.415	−0.378	0.085**	0.087**
	(0.026)	(0.269)	(0.357)	(0.028)	(0.026)
Male (%)	−0.046***	−0.047***	−0.036**	−0.042***	−0.046***
	(0.007)	(0.007)	(0.012)	(0.008)	(0.008)
Immigrant (%)	−0.002*	−0.002	−0.002*	−0.002*	−0.002*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Married (%)	0.003	0.002	−0.001	0.003	0.003
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Employment Rate (%)	−0.002	−0.02	−0.005	−0.003	−0.002
	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)
Labour Force (%)	−0.004	−0.004	−0.010***	−0.004	−0.004
	(0.002)	(0.002)	(0.003)	(0.002)	(0.003)
Average AQHI: Medium Income	−0.035***	−0.033***	—	−0.032***	−0.036***
	(0.008)	(0.008)		(0.008)	(0.009)
Average AQHI: High Income	−0.064***	−0.065***	—	−0.060***	−0.064***
	(0.008)	(0.008)		(0.008)	(0.008)
(Average AQHI)^2	—	0.123	0.093	—	—
		(0.064)	(0.082)		
Land Area (km^2)	[+]	[+]	[−]	—	[+]
Grouped by Census Designation	—	—	—	—	✓
R^2	0.802	0.812	0.632	0.793	0.802
Adjusted R^2	0.776	0.764	0.577	0.762	0.760

Note: Robust standard errors are listed in parentheses. Coefficient estimate significance is indicated as follows: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table presents the results of OLS regressions estimating the relationship between Average AQHI and Gini Coefficient across Canadian Census Metropolitan Areas (CMAs). Average AQHI is the independent variable of interest in all models. The main functional form, shown in Equation (1), includes an interaction variable between Average AQHI and CMA Mean Income. Alternative specifications (2–5) test model robustness. Alt. 1 is the functional form described in Equation (2) which adds (Average AQHI)^2. Alt. 2 is the functional form described in Equation (3) which replaces the interaction variable between Average AQHI and CMA Mean Income with (Average AQHI)^2. Alt. 3 is the functional form described in Equation (4) which omits the Land Area control variable. Alt 4. is the functional form described in Equation (5) which groups CMAs by Census Designation. The coefficient estimate β_8 for Land Area is non-significant with a magnitude $|\beta_8| < 10^{-5}$ in all functional forms.

Source: Environment and Climate Change Canada, 2025; Statistics Canada, 2021a; Statistics Canada, 2021b; Environics Analytics, 2024.

inequality.

VI. Specification Check

VII. Robustness Analysis

8. Robustness Analysis To further scrutinize the consistency of the estimated effects, an additional robustness check uses quantile regression ($\tau = 0.5$):

Quantile regression is less sensitive to outliers and allows us to see whether relationships differ at various points of the conditional Gini distribution. By focusing on the median of Gini rather than the mean, we observe whether the AQHI effect remains robust in the middle segment of the distribution. If the AQHI coefficients in the quantile regression remain significant and in the same direction as in the OLS models, this reaffirms that the relationship is not merely driven by outliers or skewed distributions in either extreme (e.g., extremely unequal or extremely equal CMAs).

While the point estimate of average AQHI shifts modestly (around -0.40), its sign remains negative, and the squared term remains positive ($+0.09$). The male (-0.053) and labor force (-0.010) variables also retain their negative associations with Gini at the median, reinforcing the baseline findings.

Across these robustness checks, the data consistently suggest that air quality's impact on income inequality is characterized by a nonlinear pattern. At lower to moderate levels of pollution, incremental increases in AQHI may correlate weakly or negatively with inequality, but once pollution surpasses a certain threshold, further deterioration appears to aggravate inequality.

VIII. Conclusion

Drawing on multiple OLS regressions and additional robustness tests, this study demonstrates a significant, nonlinear association between air quality (average AQHI) and income inequality (Gini Coefficient) within Canadian CMAs. Specifically, while mildly worse air quality does not robustly increase inequality—and may even slightly decrease it—the positive coefficients on (average AQHI)^2 indicate that poor air quality eventually aligns with rising inequality when AQHI becomes sufficiently high.

Moreover, the results highlight key demographic drivers of inequality, with higher proportions of males and immigrants predicting lower Gini, and broad labor force participation further reducing inequality. Importantly, these relationships hold under varying model specifications, including those controlling for income categories and spatial factors, as well as a quantile regression that evaluates median outcomes rather than mean outcomes.

From a policy perspective, these findings imply that environmental health interventions in highly polluted CMAs could help mitigate income inequality. Although the cross-sectional nature of the data does not allow for definitive causal claims, the U-shaped association underscores that preventing severe pollution levels may carry an additional equity benefit. Future work could employ panel data or exogenous policy shocks to better establish causality and identify the mechanisms through which poor air quality contributes to heightened inequality.

IX. References

References here (manual or bibTeX). If you are using bibTeX, add your bib file name in place of BibFile in the bibliography command.

A: Mathematical Analysis

β_1